

Fort Hays State University

2025S_CSCI441_VA_Software Engineering

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Software Engineering Project Report

Triple Tactics

Group F

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URL:

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Work Assignment

- Angel Escalera: Make and ensure documentation and prototyping of assignment information systems.
- Iain Fehr: Make and ensure a project timeline, ensure upkeep of git repository, ensure all members have proper goals and can work together effectively, and work to plan and create documentation for relevant systems and mechanics
- Juan Enriquez: Responsible for checking over assignments submissions, and idea prototyping
- SethTharo: Collaborate on the project report, ensuring all necessary sections are written and formatted properly. Take responsibility for hosting and uploading the game to the web for public access, with any final checks or adjustments handled before publication.
- Sinan: Responsible for general design suggestions and assignment quality assurance

Part 1:

Customer Problem Statement of Requirements

Problem Statement

Imagine a world where the daily grind feels like a monotonous loop: wake up, work, go home, sleep. Your phone, a gateway to endless possibilities, sits idle, offering no escape from the humdrum routine. You yearn for a spark of excitement, a quick thrill to break the monotony. Remember those carefree high school days filled with laughter and friendly competition over card games like Triple Triad? Those memories feel like a distant dream now, lost in the whirlwind of adulthood.

But what if you could recapture that magic?

Group F is bringing back the beloved card game Triple Triad, reimagined for the modern world. Introducing Triple Tactics: your pocket-sized portal to quick, engaging card battles. Picture this: you're on a short break, waiting in line, or just winding down before bed. Instead of scrolling aimlessly, you pull out your phone or tablet and dive into a fast-paced Triple Tactics match. Each game is designed to last no more than 45 seconds, making it the perfect bite-sized entertainment for those fleeting moments of free time.

Triple Tactics offers a simple, intuitive rule set that can be learned in under two minutes. Whether you prefer to read the instructions or jump right into the action, you'll be mastering the basics in no time. And the fun doesn't stop there! We're building a future for Triple Tactics that includes exciting collectability features, adding depth and replayability to the game.

Triple Tactics isn't just a game; it's a way to transform those empty moments of time into fun and friendly and, most importantly, quick competitive moments. It's a way to reconnect with the joy of card games and inject a little bit of excitement back into your day. So, ditch the mindless scrolling and embrace the quick, strategic thrills of Triple Tactics. Your daily routine deserves a little bit of magic. Tired of the daily grind? Your phone, a gateway to endless possibilities, sits idle, offering no escape from the monotony. Remember those carefree high school days, filled with laughter and friendly competition over card games like Yu-gi-oh, Magic the Gathering, and the ever-classic poker? We're bringing that magic back with Triple Tactics!

Triple Tactics is the beloved card game Triple Triad from the Final Fantasy series, reimagined for quick entertainment. It's your pocket-sized portal to quick, engaging card battles. On a short break, waiting in line, or just winding down before bed? Instead of scrolling aimlessly, dive into a fast-paced Triple Tactics match. Each game is designed to last no more than 45 seconds, making it the perfect entertainment for those fleeting moments of free time.

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Triple Tactics isn't just a game; it's a way to transform those empty pockets of time into fun and friendly competition. It's a way to reconnect with the joy of card games

and inject a little bit of excitement back into your day. So, ditch the mindless scrolling and embrace the quick, strategic thrills of Triple Tactics!

Decomposition into Sub-problems

- **Card Mechanics:** The basic rules of the card game mechanics (cards, power, location, etc.)
- **User Interface Design:** Simple and intuitive interface that fits within the quick time frame of play.
- **Game Flow:** Seamless transitions between turns, tracking of game state, and win conditions.
- **Collectability System:** Allowing players to collect and trade cards over time.
- **Multiplayer Functionality:** Connecting players for remote play with minimal lag.

Example of a gameboard design layout with each X being a card

```
+---+---+---+
| X | X | X |
+---+---+---+
| X | X | X |
+---+---+---+
| X | X | X |
+---+---+---+
```

Glossary of Terms

- **Deck:** The pool of cards not yet drawn into your hand, built before the match.
- **Hand:** The cards you currently hold are available for play.

- **Cards in play:** The cards are placed on the board.
 - **Color:** Ownership of the cards on the board, where Player One is Red and Player Two is Blue.
 - **Power:** The strength of the side of a card when interacting with another card.
 - **Location:** The position of a card on the 3x3 grid, occupying one of the nine available slots.
 - **Ruleset:** A set of instructions that governs how the game is played, such as card placement, card ownership, and winning conditions.
-

Goals, Requirements, and Analysis

Business Goals:

- **Quick and Engaging Gameplay:** The primary attraction of *Triple Tactics* is that each game can be played in under 45 seconds. This makes it ideal for casual gaming during brief moments of downtime, which appeals to a broad audience—whether they're commuting, waiting for an appointment, or taking a break at work. This quick play style encourages repeated sessions, leading to high user retention.
- **Collectibility and In-game Rewards:** *Triple Tactics* offers a collectible card system similar to popular games like *Hearthstone* or *Gacha* systems, where players can collect rare cards, trade, or even sell them in the future. This element adds long-term value for players, keeping them engaged over time. Investors will

find this appealing, as collectibility can lead to microtransactions and in-game purchases, a proven strategy for monetization in mobile games.

- **Potential for Expansion:** The game's core mechanics allow for frequent updates, including new cards, game modes, and challenges. This constant refresh keeps players engaged and provides opportunities for ongoing revenue generation through app updates, DLCs, or events.
- **Fun, Short, and Collectible:** Provide a quick, easy-to-learn game that players can enjoy in short bursts while offering opportunities for card collection.
- **Accessible Anywhere:** Enable gameplay on mobile devices with low setup time and minimal learning curve.
- **Engagement:** Include features that encourage repeat play and in-game purchases or rewards (if applicable).

Enumerated Functional Requirements

- **Must-Have:**
 - **REQ-1:** Players must be able to start a game quickly with minimal setup.
 - **REQ-2:** The game must track and update the ownership of cards dynamically based on power comparisons.
- **Nice-to-Have:**
 - **REQ-3:** Players should be able to play and end a game within 45 seconds.

Enumerated Nonfunctional Requirements

- **Must-Have:**
 - **Performance:** The game should run smoothly on any chosen platform, be it Android, iOS, or Windows devices, with minimal lag.
- **Nice-to-Have:**
 - **Scalability:** Future updates should allow for the addition of more cards and features without overhauling the system.

User Interface Requirements

- The user interface must present the 3x3 grid clearly, with each player's hand easily accessible.
 - Players must be able to select and place cards in available slots quickly.
 - The game should provide clear feedback on ownership changes (e.g., color changes on cards) and display the winner at the end of the match.
-

Part 2:

Functional Requirement Specification and Use Cases

Stakeholders

- **Players (End Users)** - Individuals who engage with the game for entertainment.
- **Game Developers** - The development team responsible for coding and maintaining the game.

- **Game Moderators** - Ensuring fair play and managing in-game reports.

Actors and Goals

Initiating Actors:

1. Player:

- Start a new game.
- Place cards on the board.
- Competes against an opponent.
- Collects and trades cards.
- View their card collection.
- Goal: Engage in and win matches.

Participating Actors:

1. Game System:

- Manages game state.
- Determines win conditions.
- Updates player statistics.
- Provides real-time feedback (e.g., color changes in ownership).

2. Opponent (Computer/Bot):

- Competes against the player in the match.

Use Cases

Casual Description

1. Start a New Game

- a. Description: A player selects the "New Game" option to start a match.
- b. Responds to Requirements: REQ-1, REQ-3

2. Place a Card

- a. Description: A player selects a card from their hand and places it on the board.
- b. Responds to Requirements: REQ-2

3. Determine Game Winner

- a. Description: At the end of the game, the system calculates the winner based on card ownership.
- b. Responds to Requirements: REQ-2

4. Collect and Trade Cards

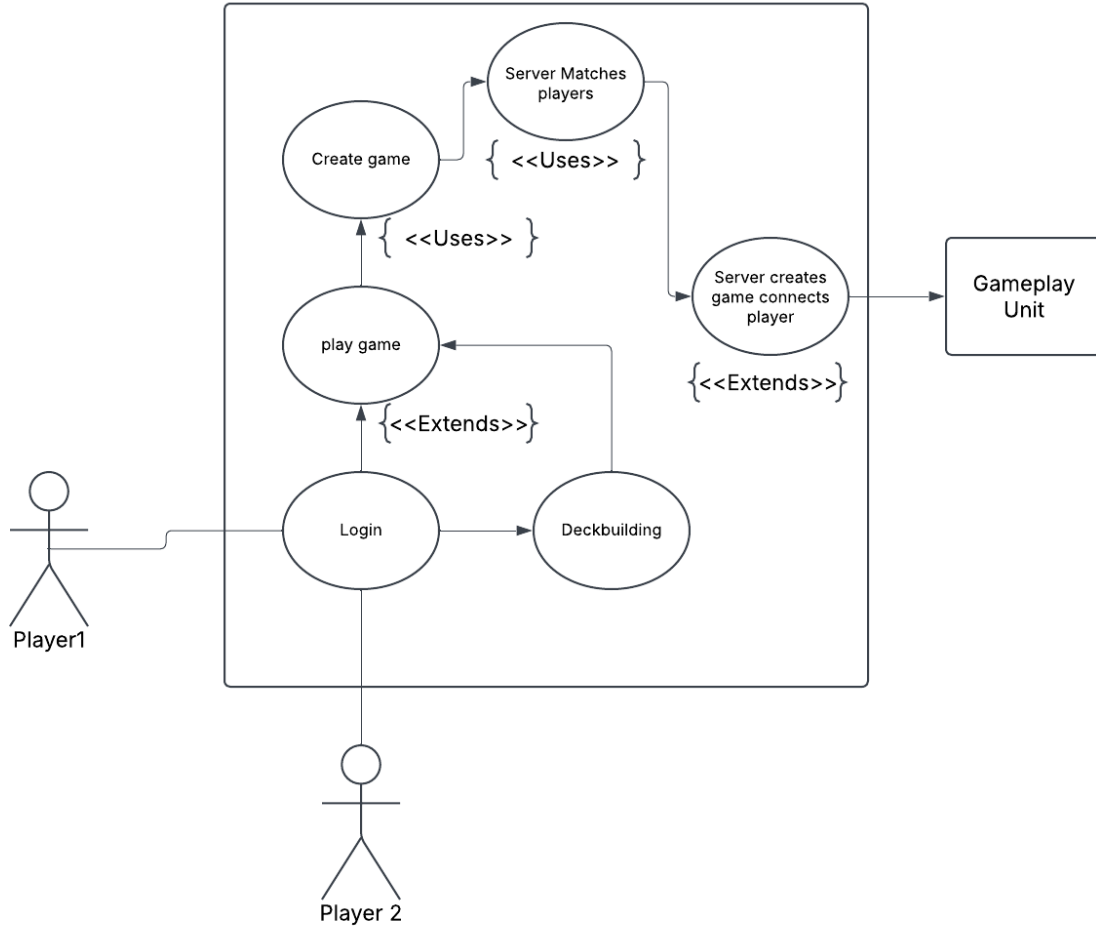
- a. Description: Players can collect new cards and trade with others to build their deck.
- b. Responds to Requirements: Functional Collectibility System

5. View Card Collection

- a. Description: Players can browse their owned cards and manage their deck.
- b. Responds to Requirements: User Interface Design

Base use case Diagram

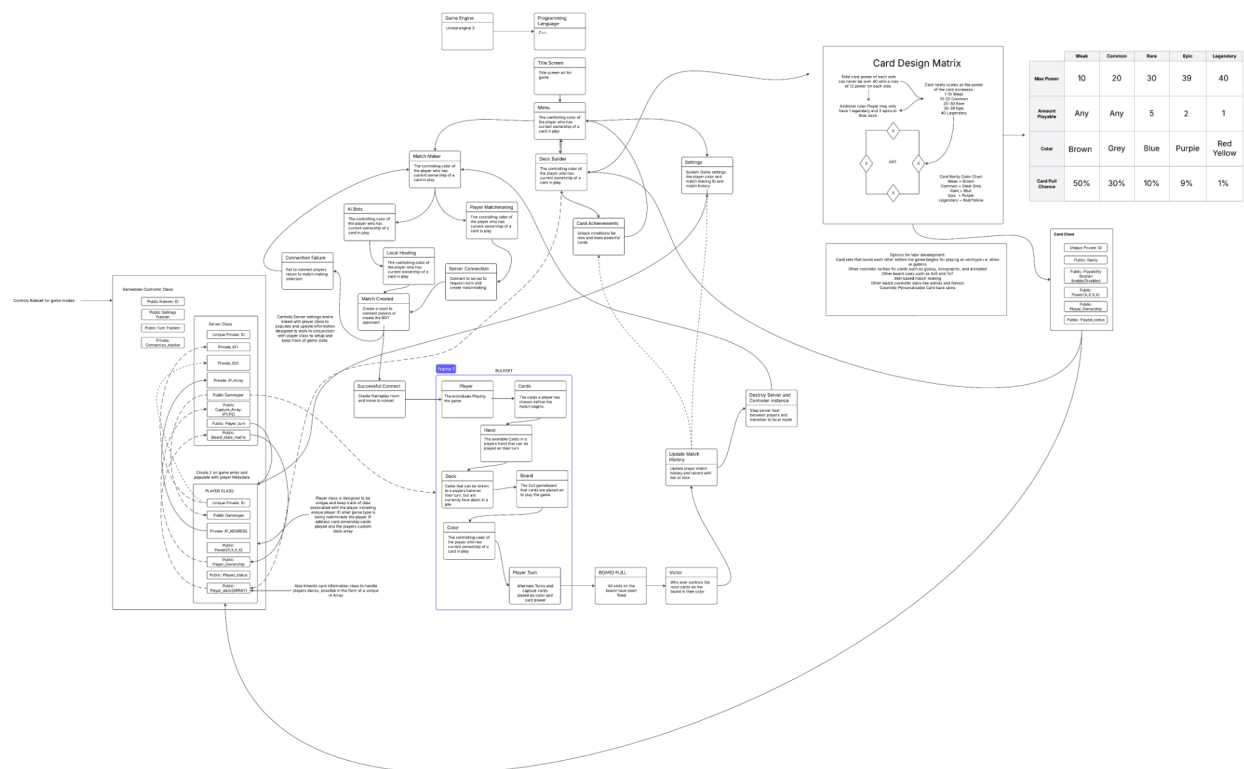
Game Use Case



Traceability Matrix

	A	B	C	D	E	F
1	Use Case	Requirement	Priority	Designed	Testing	Deployment
2	Login	Creates and holds player unique ID and information	0	No	Not started	Not started
3	Play Game	Menu system for player choice starts matchmaking process	1	No	Not started	Not started
4	Deck_builder	Deck builder for player progression and deck choice	2	No	Not started	Not started
5	Create Game	Essential component starts the create game process	1	In-Progress	Not started	Not started
6	Server_matching	Essential component contacts server and executes player match making	1	No	Not started	Not started
7	game_creation	Essential component creates gameplay room for players to interact	1	No	Not started	Not started
8	gameplay_unit	Essential component room for player connection and interaction	0	No	Not started	Not started
9						

TOTAL Gamedesign Diagram



Full-Dressed Description

A. Use Case: Start a New Game

Actors: Player, Game System

Preconditions: Player has launched the game and is on the main menu.

Trigger: Player selects "New Game."

Basic Flow:

1. The player selects "New Game."
2. The system displays the 3x3 grid and player's hand.
3. The game begins with a bot player starting.
4. The system waits for the player's first move.

Postconditions: The game state is initialized, and the first move is expected.

B. Use Case: Place a Card

Actors: Player, Game System

Preconditions: The player has started a game, and it is their turn.

Trigger: Player selects a card and a position on the board.

Basic Flow:

1. The player selects a card from their hand.
2. The player taps a valid position on the board.
3. The system places the card and evaluates ownership changes.
4. The system updates the board state and shifts turns.

Postconditions: The game state updates with the new card placement.

System Sequence Diagram

The user interface for *Triple Tactics* is designed for quick, intuitive gameplay with minimal navigation effort. Below is a step-by-step breakdown of the UI flow for key use cases.

Start a New Game

1. **Main Menu:** The player sees three options:
 - **New Game**
 - **Card Collection**
 - **Settings**
2. **Game Initialization:** Upon selecting “New Game,” the screen transitions to a 3x3 grid game board. The player's hand appears at the bottom.
3. **Match Start:** A prompt displays, indicating if the player or the AI goes first.

Place a Card

1. The player taps a card from their hand.
2. The available slots highlight valid placements.
3. The player taps a slot to place the card.
4. The system updates the board, evaluates ownership changes, and shifts the turn to the AI.
5. The AI makes its move automatically.

Determine Game Winner

1. Once all nine slots are filled, the system evaluates the board based on card ownership.

2. A message displays the winner and the final board state.
3. The player sees a summary screen with the option to start a new game or return to the main menu.

Collect and View Cards

1. The player navigates to the “Card Collection” menu.
2. They can browse, filter, and inspect individual cards.

User Effort Estimation

Start a New Game

- **Clicks/Taps:** 2 (Main Menu → New Game)
- **Navigation Effort:** Low

Place a Card

- **Clicks/Taps:** 3 (Select Card → Choose Slot → Confirm Placement)
- **Navigation Effort:** Low

Determine Game Winner

- **Clicks/Taps:** 1 (Acknowledge Results)
- **Navigation Effort:** Minimal

View Card Collection

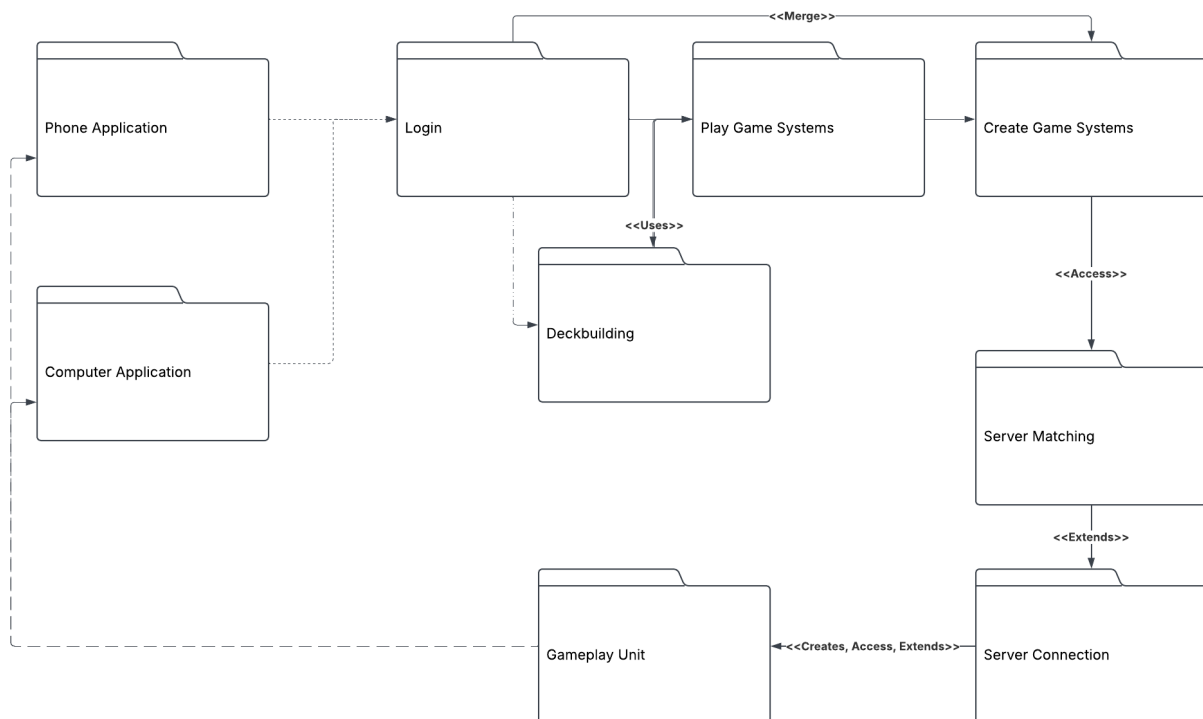
- **Clicks/Taps:** 2+ (Navigate to Collection → Browse Cards)

- **Navigation Effort:** Low

Part 3:

System Architecture and System Design

1. Identifying Subsystems via UML Chart



2. Architecture Styles

This project's main architectural style is layering. As a modular-based system its extremely important that every system can interconnect and be used in the pipeline by the main gameplay process. As such, each layer is designed if a

specific set of functions that will execute then pass down the correct information to the next layer in the pipeline to either expand or modify as needed. The primary architectural style of this project is layering. As a modular-based system, it is imperative that every system can interconnect and be utilized within the pipeline by the main gameplay process. Therefore, each layer is designed with a specific set of functions that will execute and then transfer the necessary information to the subsequent layer in the pipeline, to be expanded or modified as needed.

3. Mapping Subsystems to Hardware

In the final build, the game will run both on the client and on the server. The game client will update the server and prevent IP address direction connections, acting like a mask for player security. However, it is also possible that a server is unnecessary, and a direct IP connection will be formed between 2 client devices.

4. Connectors and Network Protocols

If a server-side connection is made, it will be programmed using Javascript and React using a REST API framework to ensure that the game can be updated in an expedient manner to prevent any limitations on player interaction with the game state, ensuring a smooth and reliable interaction for players.

5. Global Control Flow

- **Execution orders:** The gameplay systems are designed to be linear as a turn-based game. The gameplay is designed to be executed and uploaded to the server to update the other player's game client and then repeated by the other player.

- **Time dependency:** This system will be focused on an event-based system with an included timeout. Normal gameplay will be designed to progress from player to player after each player plays or performs an action on their turn. A timeout will also be included to prevent players from leaving the game session, preventing players from quitting and leaving the other player within a one-person game. This timeout will result in an automatic loss of the player's timing out.

6. Hardware Requirements

Color display, with a minimum resolution of 1280×1080 pixels.

Minimum of 3 Gigs of hard disk space.

Internet connection is required for online play.

Minimum 4 gigs of RAM

Touch or User interface device required for gameplay

6.

Plan of Work: Project Timeline

	February				March				April				May			
	1-8	8-15	15-22	22-28	1-7	8-15	16-22	23-29	30-4	5-11	12-19	20-25	26-2	2-9	10	
Iain Fehr	Learn Unreal Engine				Create 2D Game environment				Create Base Card System				Game Art and Front Design			
Angel Escalera			Learn Unreal Engine				Create Title Menu		Create Deckbuilding system		Card Creation	Gameplay Testing	Local BOT Systems	Final QA Tests	TURN IN	
Juan Enriquez			Learn Unreal Engine				Create Title Menu		Client Matchmaking REST		Gameplay Testing	Card Creation	Local BOT Systems	Final QA Tests	TURN IN	
Seth Tharo			Learn Unreal Engine				Create Base Card System		Create Deckbuilding System		Gameplay Testing	Card Creation	Local BOT Systems	Final QA Tests	TURN IN	
Sinan			Learn Unreal Engine				Create REST API for server		Client Matchmaking REST		Card Creation	Gameplay Testing	Local BOT Systems	Final QA Tests	TURN IN	

7. References

<https://dev.epicgames.com/community/learning/tutorials/e2V/your-first-game-in-unreal-engine-5>

PLAYLIST: <https://www.youtube.com/watch?v=L9qixi858Ag&list=PLIn-yd4vnXbjWeYqU7epakdnVzoysMToy>

Conclusion:

Overall, this is a reasonably large project that will need multiple fronts of development completed at a time to complete the project successfully on time. Considerations will have to be made for the development of systems as each system is essential to overall success. Each module should be easy to implement but will need to work seamlessly with any other modules to create a cohesive final product.